

JEE MAIN REVISION MASTER: STATES OF MATTER

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 8 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Gaseous State — Core Principles & Governing Laws	8 PYQs	Core Principle
Standard	Gaseous State — Applied Cases, Reactions & Traps	1 PYQs	Applied Topics
Standard	Gaseous State — Standard Calculations & Problem Archetypes	0 PYQs	Quantitative Analysis

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Gaseous State — Core Principles & Governing Laws	$(T_2)^{-1} \propto 2 \times (T_1)^{-1}$ (initial temperature), $pV = RT$
Gaseous State — Applied Cases, Reactions & Traps	$\text{HCO}_3^- \rightleftharpoons \text{CO}_3^{2-} + \text{H}^+$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Gaseous State — Core Principles & Governing Laws (High Yield — 8 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in Gaseous State.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in Gaseous State.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in Gaseous State PYQs.

Representative Source PYQs:

- [2024 Shift] Q#1 (MCQ): "Identify the wrong relation for real gases :..." [Ans: N/A]
- [2022 Shift] Q#2 (MCQ): "Which property of CO_2 makes it biologically and geo-chemically important?..." [Ans: N/A]
- [2022 Shift] Q#3 (MCQ): "The volume of 2.8g of CO at 27°C and 0.821 atm pressure is ($R=0.08210\text{ L. atm Kmol}^{-1}$)..." [Ans: N/A]

3.2 Gaseous State — Applied Cases, Reactions & Traps (Standard Yield — 1 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of Gaseous State.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- [2022 Shift] Q#2 (MCQ): "Which property of CO_2 makes it biologically and geo-chemically important?..." [Ans: N/A]

3.3 Gaseous State — Standard Calculations & Problem Archetypes (Standard Yield — 0 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in Gaseous State.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- [2024 Shift] Q#1 (MCQ): "Identify the wrong relation for real gases :..." [Ans: N/A]
- [2022 Shift] Q#2 (MCQ): "Which property of CO_2 makes it biologically and geo-chemically important?..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2022 (Q#2)

Which property of CO_2 makes it biologically and geo-chemically important?

- (A) Its colourless and odourless nature
- (B) Its low solubility in water
- (C) Its high compressibility
- (D) Its acidic nature

Correct Answer: None

Solution Analysis: The property of low solubility of carbon dioxide in water makes it biologically and geo-chemically important. Carbon dioxide on reaction with water form carbonic acid, which dissociates to give HCO_3^- ions. $\text{H}_2\text{CO}_3/\text{HCO}_3^-$ buffer system helps to maintain pH of blood between 7.26-7.42 .